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Engineering Circuit Analysis

Eighth Edition

Chapter 15 Circuit Analysis in s-Domain

Complex frequency and Laplace transform

We had complex frequency in RLC circuit $s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$

In most general case $\boxed{s = \sigma + j\omega}$

σ - **real part of complex frequency**, or **neper frequency** or simply **nepers**

$$v(t) = \operatorname{Re}[V_m e^{j\theta} e^{st}]$$

$$v(t) = V_m e^{\sigma t} \cos(\omega t + \theta)$$

$$i(t) = \operatorname{Re}[I_m e^{j\phi} e^{st}]$$

$$i(t) = I_m e^{\sigma t} \cos(\omega t + \phi)$$

Any signal can be expressed through the sum of exponentials



Laplace transform

Forward

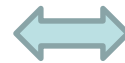
inverse

two-sided

$$\mathbf{F}(s) = \int_{-\infty}^{+\infty} e^{-st} f(t) dt$$

one-sided

$$\mathbf{F}(s) = \int_{0^-}^{+\infty} e^{-st} f(t) dt$$



$$f(t) = \frac{1}{2\pi j} \int_{\sigma_0 - j\infty}^{\sigma_0 + j\infty} e^{st} \mathbf{F}(s) ds$$

Compute Laplace transform of the functions:

$$f(t)=u(t)$$

$$f(t)=u(t)*\exp(-at)$$

$$f(t)=tu(t)$$

$$f(t)=t\exp(-at)u(t)$$

$$f(t)=\delta(t)$$

TABLE 14.1 Laplace Transform Pairs

$f(t) = \mathcal{L}^{-1} \{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$	$f(t) = \mathcal{L}^{-1} \{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1	$\frac{1}{\beta - \alpha} (e^{-\alpha t} - e^{-\beta t}) u(t)$	$\frac{1}{(s + \alpha)(s + \beta)}$
$u(t)$	$\frac{1}{s}$	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
$tu(t)$	$\frac{1}{s^2}$	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$
$\frac{t^{n-1}}{(n-1)!} u(t), n = 1, 2, \dots$	$\frac{1}{s^n}$	$\sin(\omega t + \theta) u(t)$	$\frac{s \sin \theta + \omega \cos \theta}{s^2 + \omega^2}$
$e^{-\alpha t} u(t)$	$\frac{1}{s + \alpha}$	$\cos(\omega t + \theta) u(t)$	$\frac{s \cos \theta - \omega \sin \theta}{s^2 + \omega^2}$
$te^{-\alpha t} u(t)$	$\frac{1}{(s + \alpha)^2}$	$e^{-\alpha t} \sin \omega t u(t)$	$\frac{\omega}{(s + \alpha)^2 + \omega^2}$
$\frac{t^{n-1}}{(n-1)!} e^{-\alpha t} u(t), n = 1, 2, \dots$	$\frac{1}{(s + \alpha)^n}$	$e^{-\alpha t} \cos \omega t u(t)$	$\frac{s + \alpha}{(s + \alpha)^2 + \omega^2}$

TABLE 14.2 Laplace Transform Operations

Operation	$f(t)$	$F(s)$
Addition	$f_1(t) \pm f_2(t)$	$F_1(s) \pm F_2(s)$
Scalar multiplication	$kf(t)$	$kF(s)$
Time differentiation	$\frac{df}{dt}$	$sF(s) - f(0^-)$
	$\frac{d^2f}{dt^2}$	$s^2F(s) - sf(0^-) - f'(0^-)$
	$\frac{d^3f}{dt^3}$	$s^3F(s) - s^2f(0^-) - sf'(0^-) - f''(0^-)$
Time integration	$\int_{0^-}^t f(t) dt$	$\frac{1}{s}F(s)$
	$\int_{-\infty}^t f(t) dt$	$\frac{1}{s}F(s) + \frac{1}{s} \int_{-\infty}^{0^-} f(t) dt$
Convolution	$f_1(t) * f_2(t)$	$F_1(s)F_2(s)$
Time shift	$f(t-a)u(t-a), a \geq 0$	$e^{-as}F(s)$
Frequency shift	$f(t)e^{-at}$	$F(s+a)$
Frequency differentiation	$tf(t)$	$-\frac{dF(s)}{ds}$
Frequency integration	$\frac{f(t)}{t}$	$\int_s^\infty F(s) ds$
Scaling	$f(at), a \geq 0$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Initial value	$f(0^+)$	$\lim_{s \rightarrow \infty} sF(s)$
Final value	$f(\infty)$	$\lim_{s \rightarrow 0} sF(s), \text{ all poles of } sF(s) \text{ in LHP}$
Time periodicity	$f(t) = f(t+nT),$ $n = 1, 2, \dots$	$\frac{1}{1-e^{-Ts}}F_1(s),$ where $F_1(s) = \int_{0^-}^T f(t)e^{-st} dt$

Calculate inverse Laplace transform of the functions:

$$F(s) = \frac{7}{s} - \frac{31}{s+17}$$

$$F(s) = \frac{2}{s} - \frac{4}{s^2} + \frac{3.5}{(s+10)(s+10)}$$

$$F(s) = 2 \frac{s+2}{s}$$

$$F(s) = \frac{3s^2 - 4}{s^2}$$

Method of Residues

$$V(s) = \frac{1}{(s+a)(s+b)} = \frac{A}{s+a} + \frac{B}{s+b}$$

$$A = (s+a) \cdot V(s) \Big|_{s=-a} = \frac{1}{b-a}$$

$$B = (s+b) \cdot V(s) \Big|_{s=-b} = \frac{1}{a-b}$$

Calculate inverse Laplace transform of the functions:

$$F(s) = \frac{7s+5}{s^2+s}$$

$$F(s) = \frac{5}{s^2 - 5s + 6}$$

$$F(s) = \frac{11s+30}{s^2+3s}$$

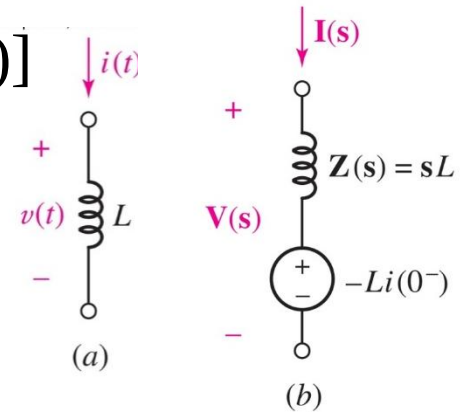
Analysis in s-domain

Resistors: $v(t) = Ri(t) \xrightarrow{L} V(s) = RI(s)$ $R = Z_R(s) = \frac{V(s)}{I(s)}$

Inductors: $v(t) = L \frac{di(t)}{dt} \xrightarrow{L} V(s) = L[sI(s) - i(0^-)]$

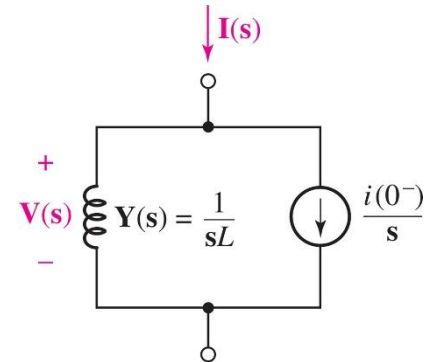
If $i(0^-) = 0$, then:

$Z_L(s) = \frac{V(s)}{I(s)} = sL$

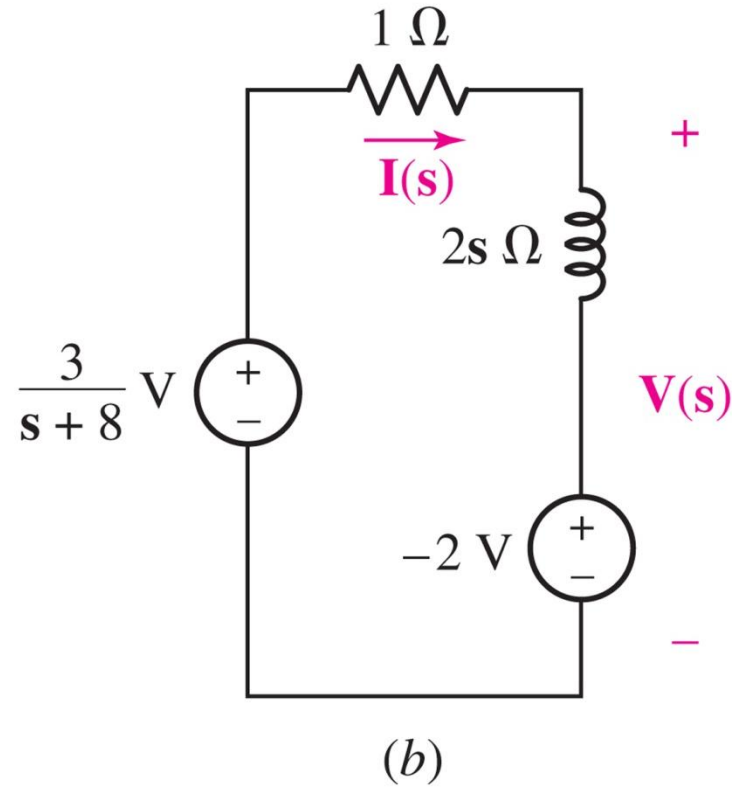
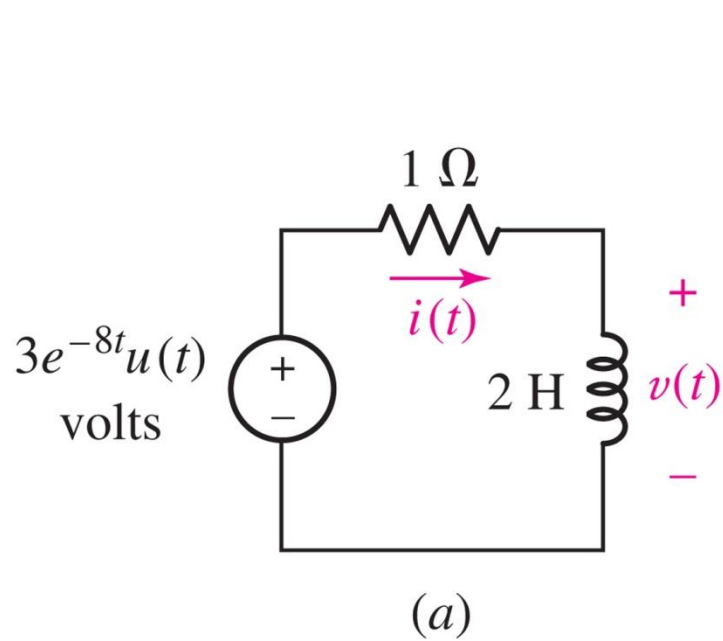


If $\sigma = 0$ (steady state response), then: $Z_L(\omega) = j\omega L$

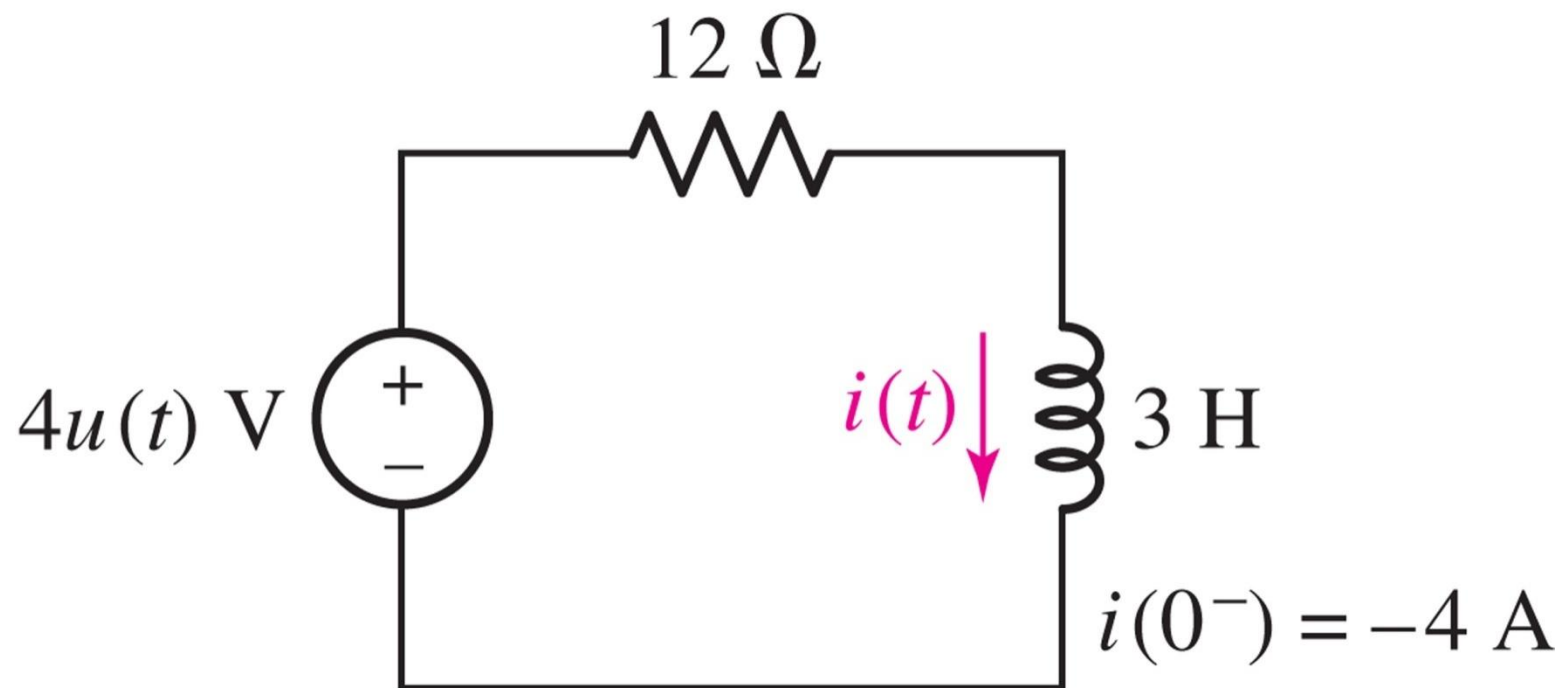
$$I(s) = \frac{V(s)}{sL} + \frac{i(0^-)}{s}$$



Calculate the voltage $v(t)$, given an initial current $i(0^-)=1$ A



Calculate the current $i(t)$, given an initial current $i(0^-) = -4$ A



$$\frac{1}{3}[1-13\exp(-4t)]u(t)$$

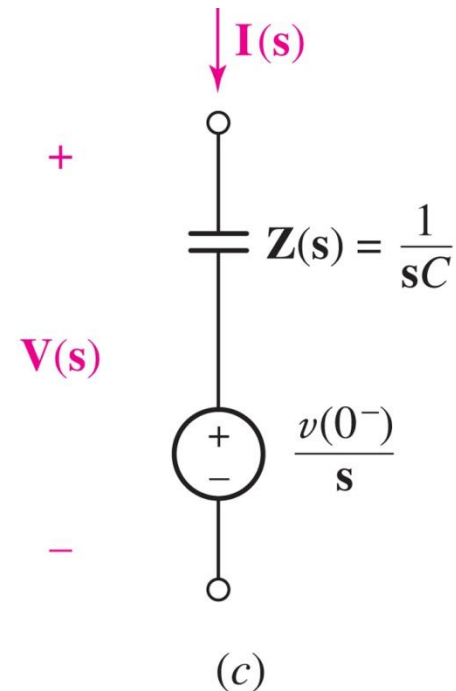
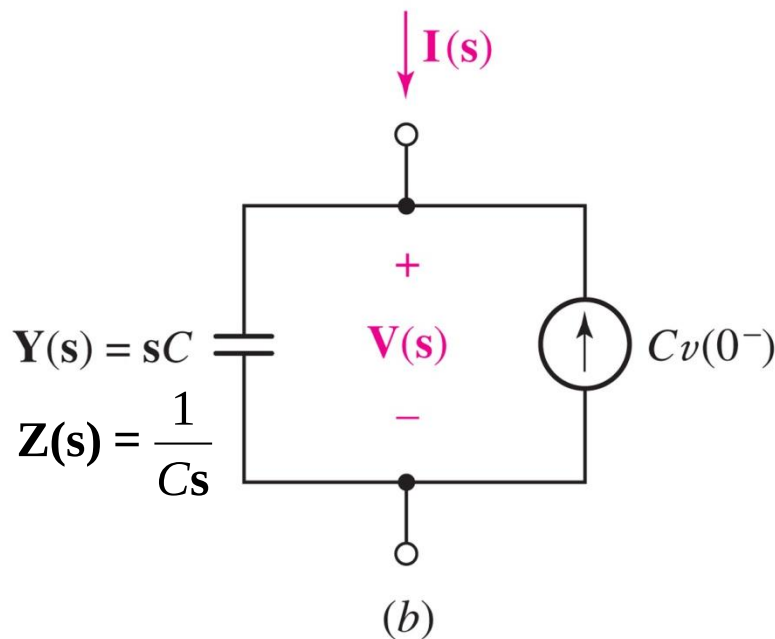
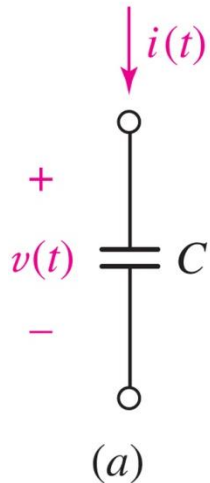
Analysis in s-domain

Capacitors: $i(t) = C \frac{dv(t)}{dt} \xrightarrow{L} \mathbf{I}(s) = C[s\mathbf{V}(s) - v(0^-)]$

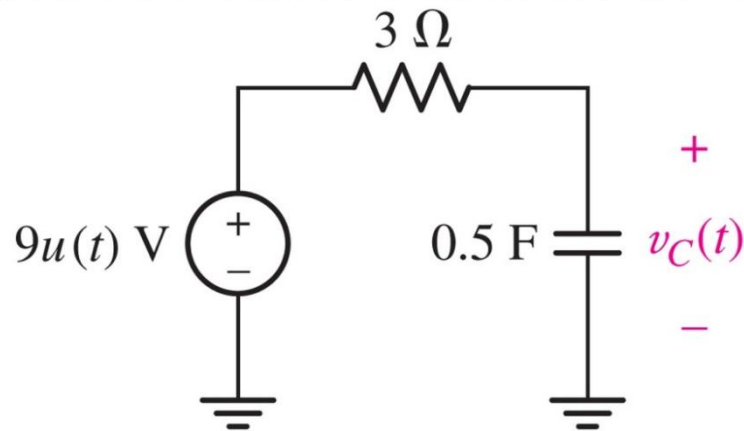
If $v(0^-) = 0$, then:

$$\mathbf{Z}_C(s) = \frac{\mathbf{V}(s)}{\mathbf{I}(s)} = \frac{1}{sC}$$

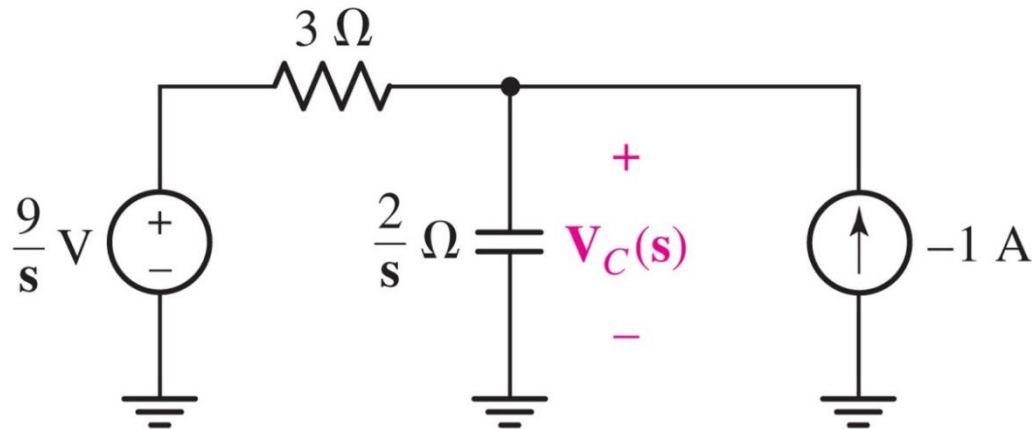
If $\sigma = 0$, then: $\mathbf{Z}_C(\omega) = \frac{1}{j\omega C}$



Determine $v_C(t)$, given an initial voltage $v_C(0^-) = -2$ V, using 2 equivalents



(a)



(b)

$$9u(t) - 11e^{-2t/3}u(t) \text{ V}$$

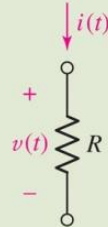
Analysis in s-domain

Time Domain

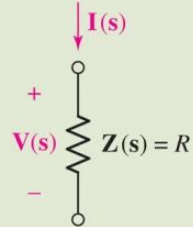
Frequency Domain

Resistor

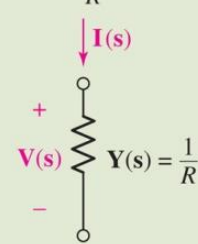
$$v(t) = Ri(t)$$



$$V(s) = RI(s)$$

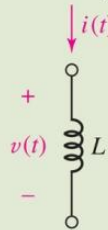


$$I(s) = \frac{1}{R} V(s)$$

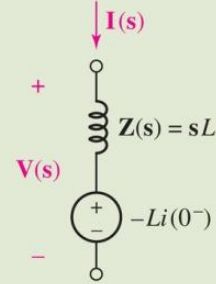


Inductor

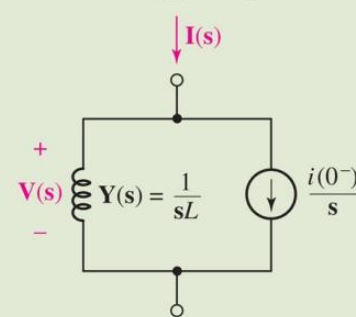
$$v(t) = L \frac{di}{dt}$$



$$V(s) = sLI(s) - Li(0^-)$$

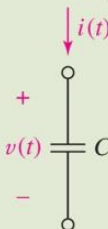


$$I(s) = \frac{V(s)}{sL} + \frac{i(0^-)}{s}$$

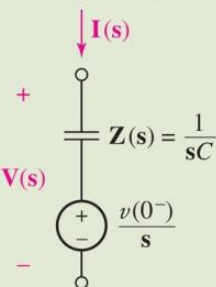


Capacitor

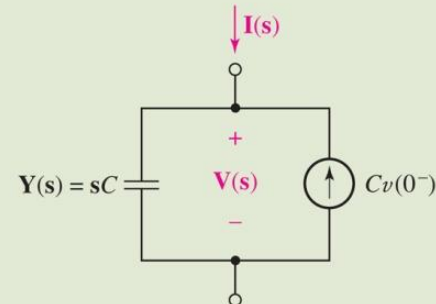
$$i(t) = C \frac{dv}{dt}$$



$$V(s) = \frac{I(s)}{sC} + \frac{v(0^-)}{s}$$

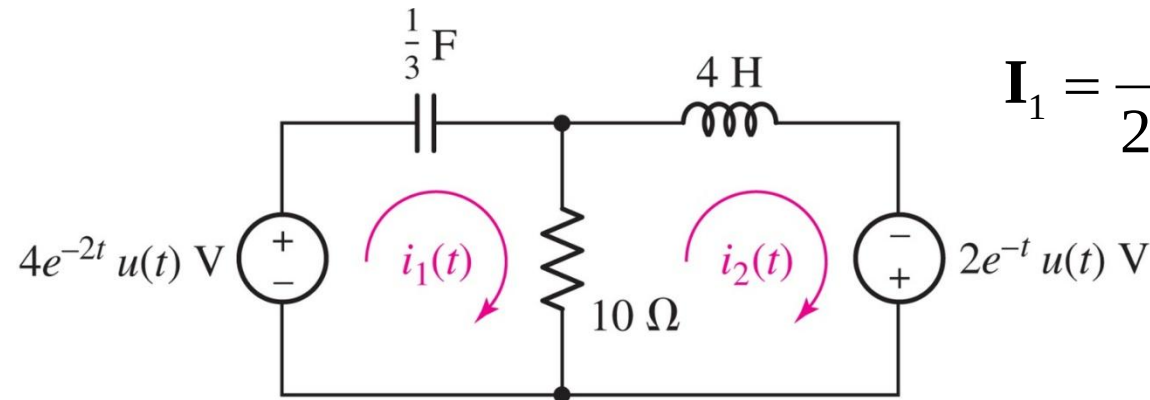


$$I(s) = sCV(s) - Cv(0^-)$$



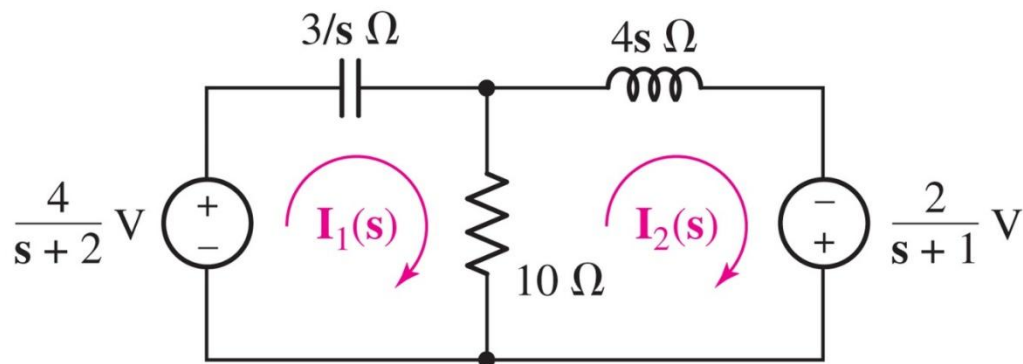
Nodal and Mesh Analysis

Determine the two mesh currents in the circuit. There is no energy initially stored in the circuit



(a)

$$\mathbf{I}_1 = \frac{2s(4s^2 + 19s + 20)}{20s^4 + 66s^3 + 73s^2 + 57s + 30}$$

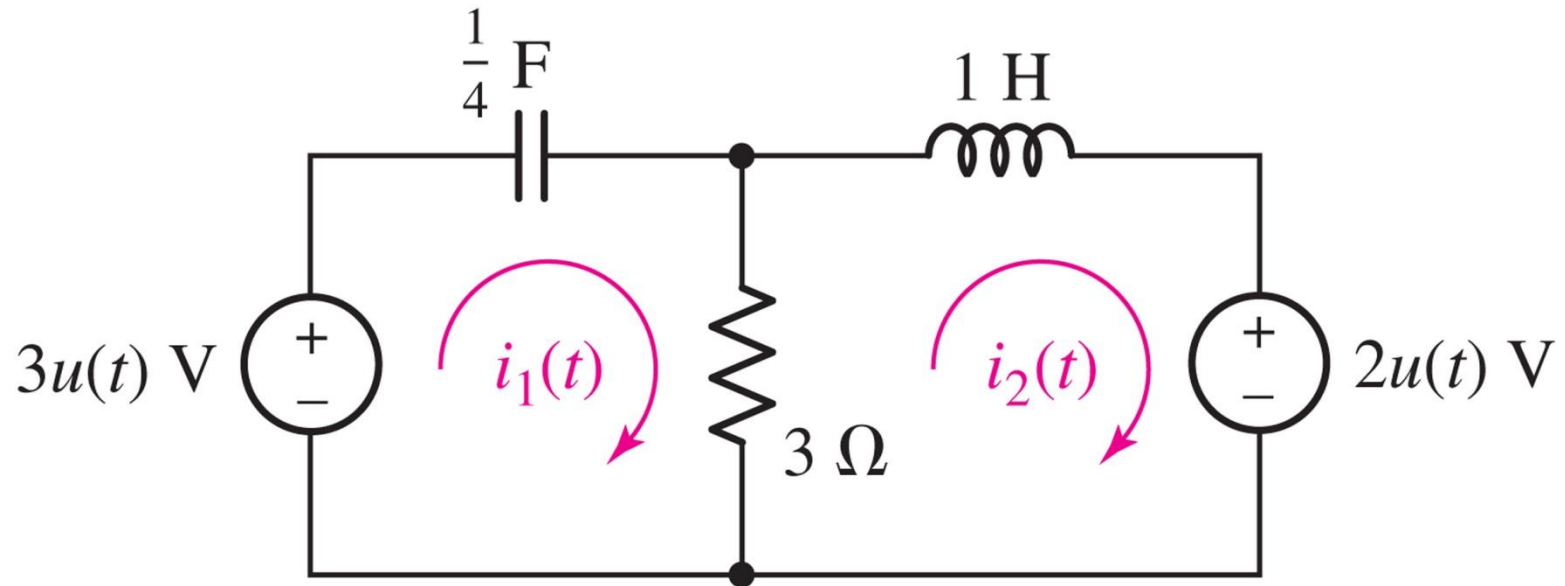


(b)

$$\mathbf{I}_2 = \frac{30s^2 + 43s + 6}{(s + 2)(20s^3 + 26s^2 + 21s + 15)}$$

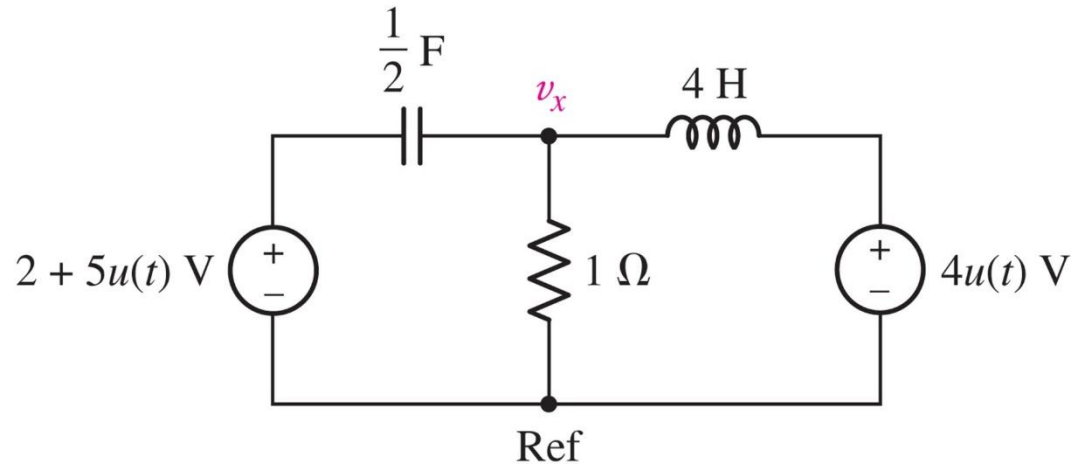
Find two mesh currents in the circuit. There is no energy initially stored in the circuit at $t=0^-$

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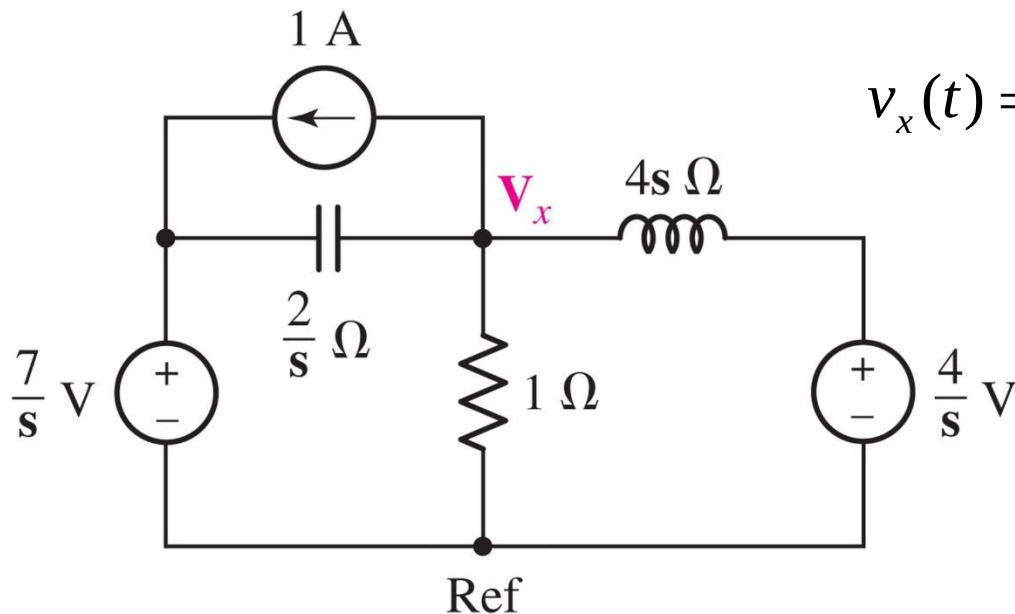
Nodal and Mesh Analysis

Find v_x using nodal analysis



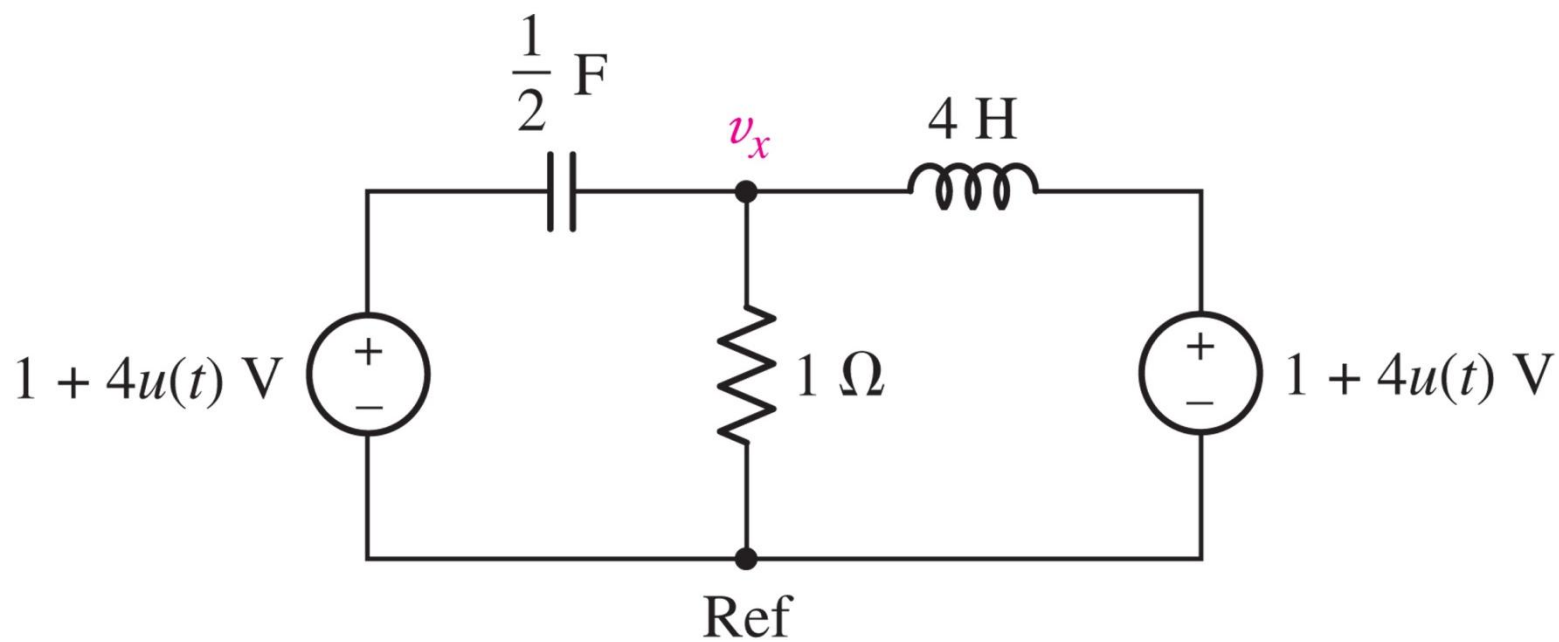
$$V_c(0) = 2V$$

$$\mathbf{V}_x = \frac{10s^2 + 4}{s(2s^2 + 4s + 1)}$$



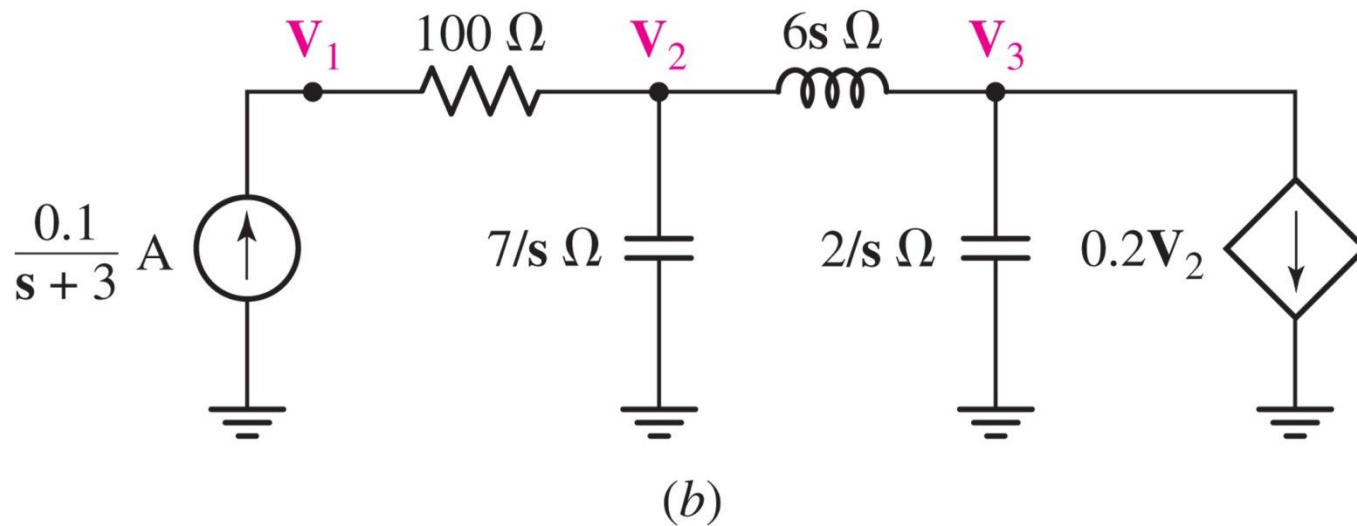
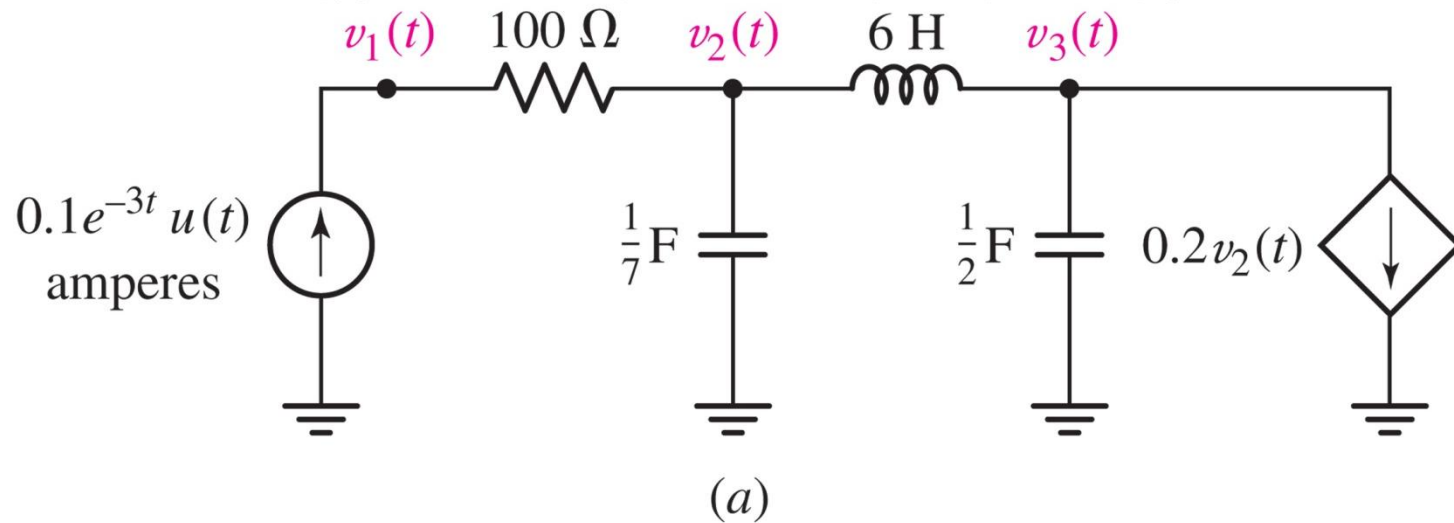
$$v_x(t) = 4 + 6.864e^{-1.707t} - 5.864e^{-0.2929t}$$

Find v_x using nodal analysis



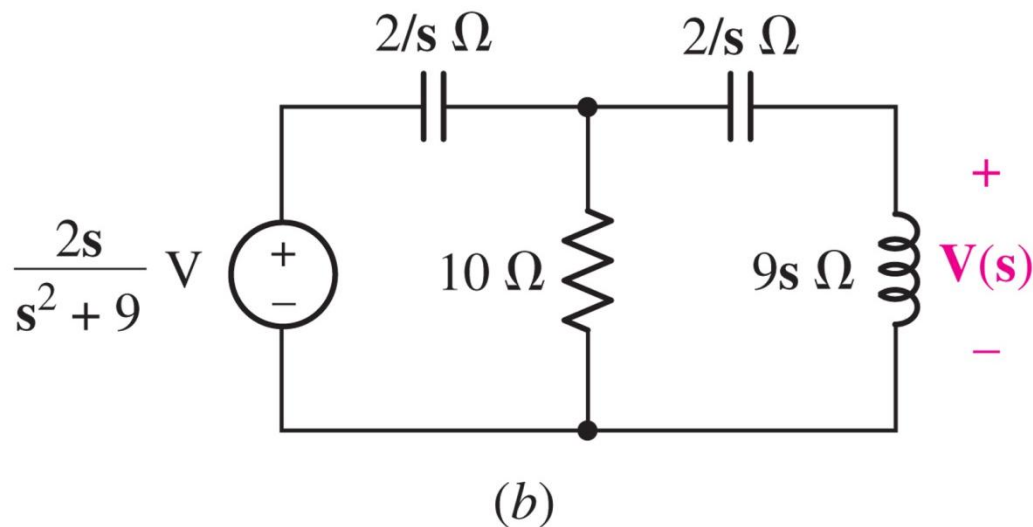
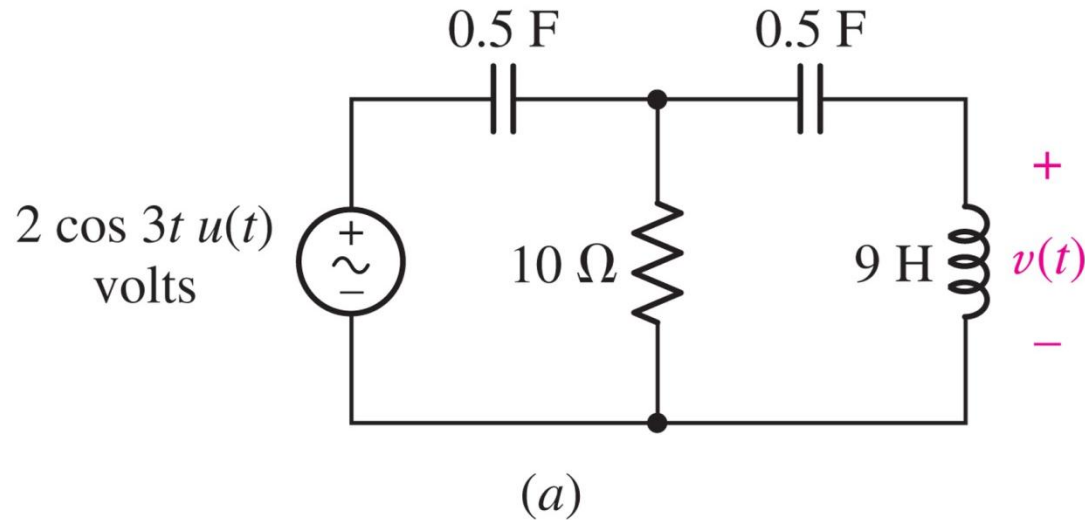
Nodal and Mesh Analysis

Find v_1 , v_2 and v_3 using nodal analysis. No initial energy is stored in the circuit



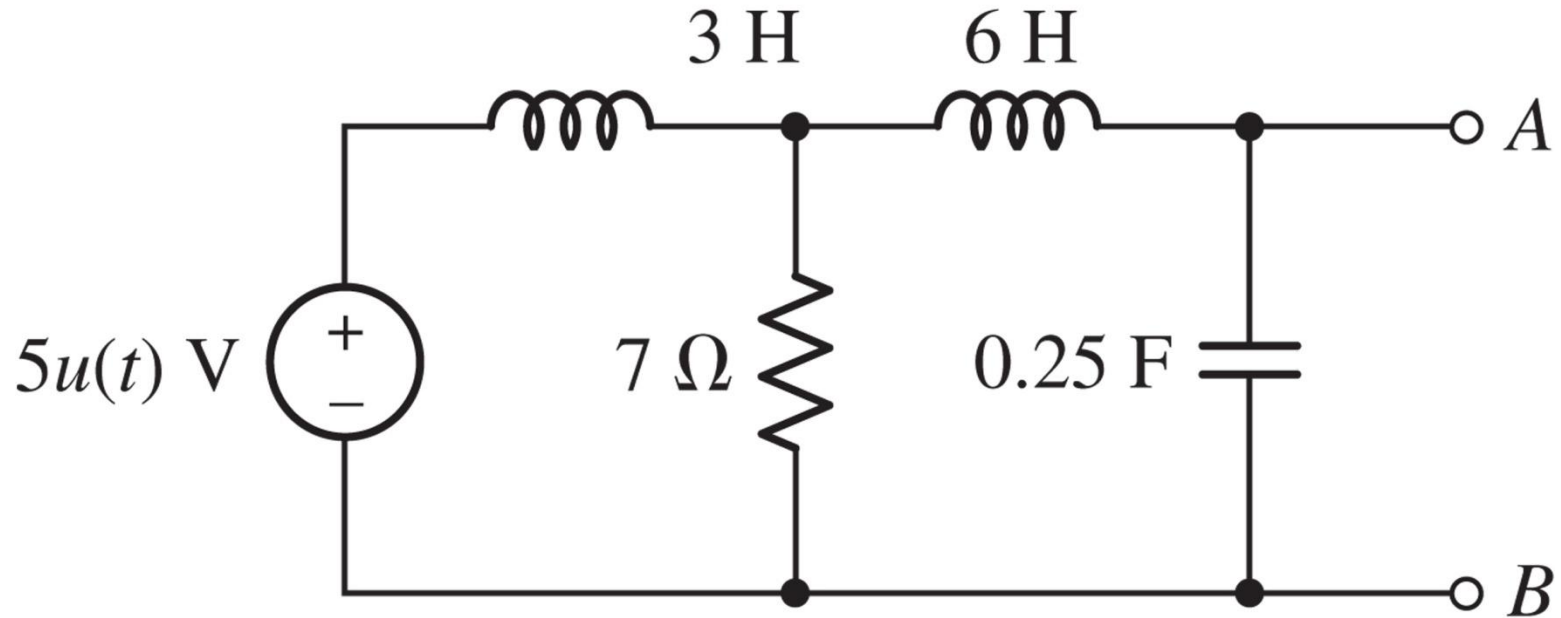
Source transformations

Simplify the circuit using source transformation technique. No initial energy stored at $t=0^-$.



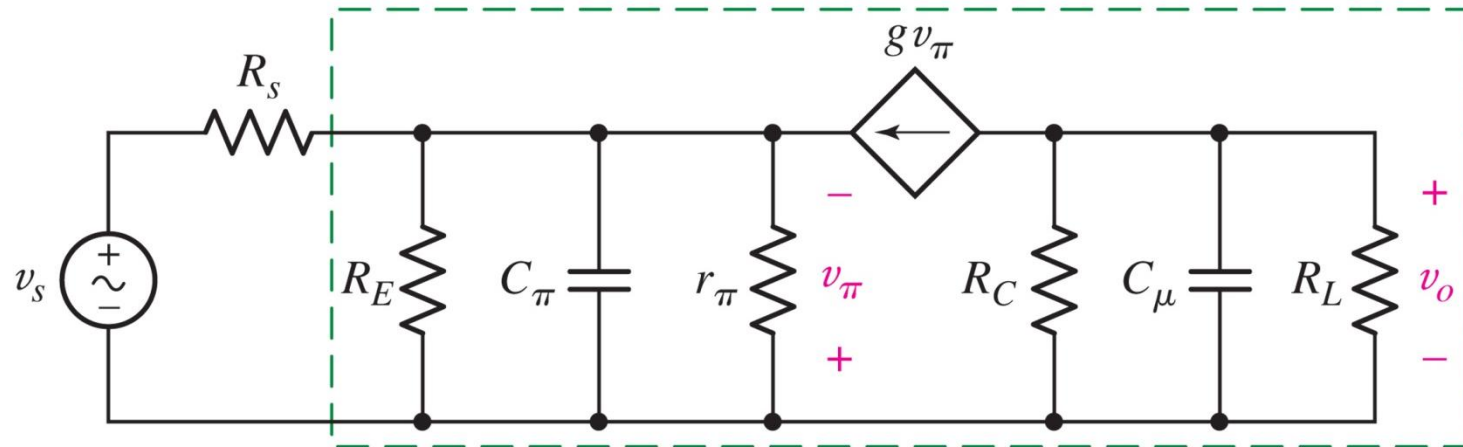
Simplify the circuit using source transformation technique. No initial energy stored at $t=0^-$.

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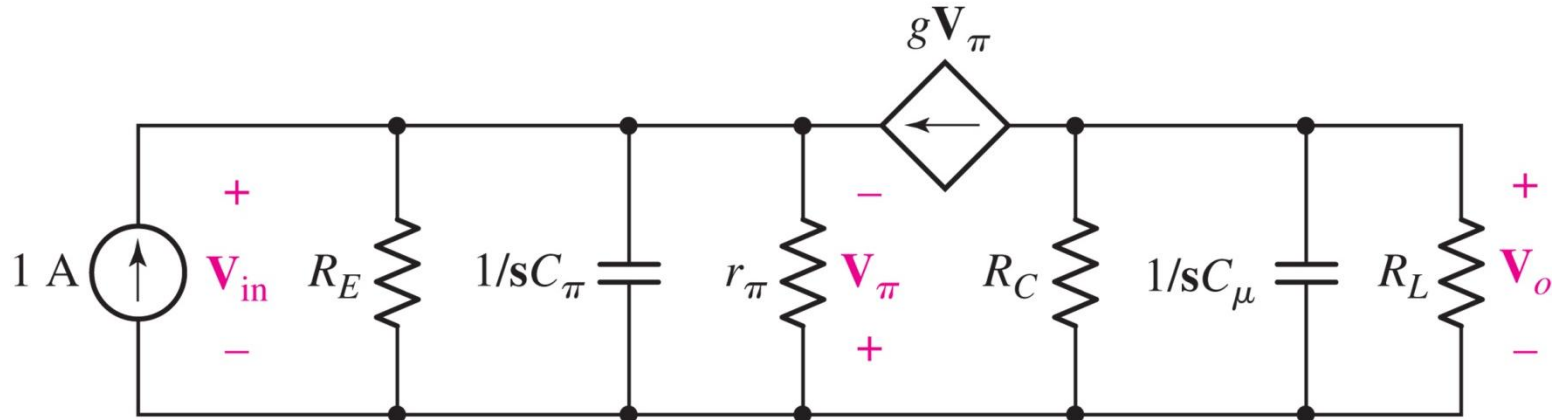


Thevenin equivalent

Find the frequency-domain Thevenin equivalent of the highlighted network.
No initial energy stored at $t=0^-$.



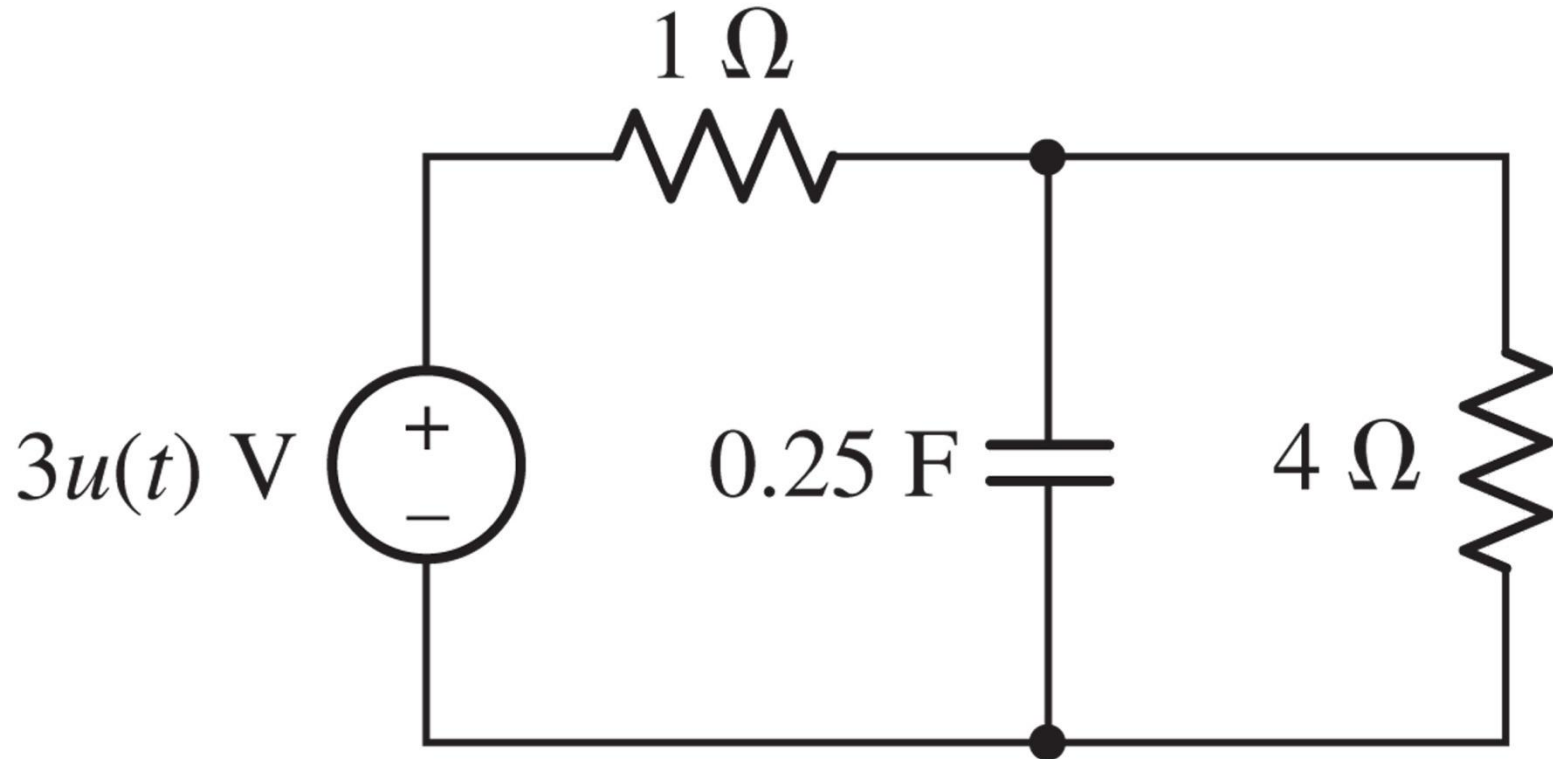
(a)



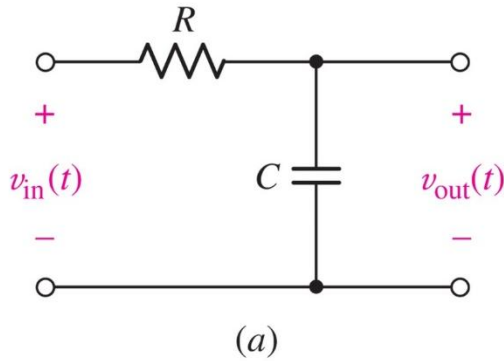
(b)

Find the s-domain Norton equivalent of the network connected to 1 Ohm resistor. No initial energy stored at $t=0^-$.

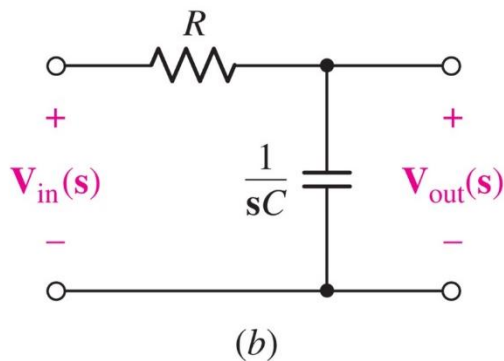
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Poles, Zeros and Transfer functions



$$\frac{V_{out}}{1/(sC)} + \frac{V_{out} - V_{in}}{R} = 0$$



$$H(s) = \frac{V_{out}}{V_{in}} = \frac{1}{1 + sRC}$$

**Transfer function
or System function**

$$s \rightarrow \infty, H \rightarrow 0$$

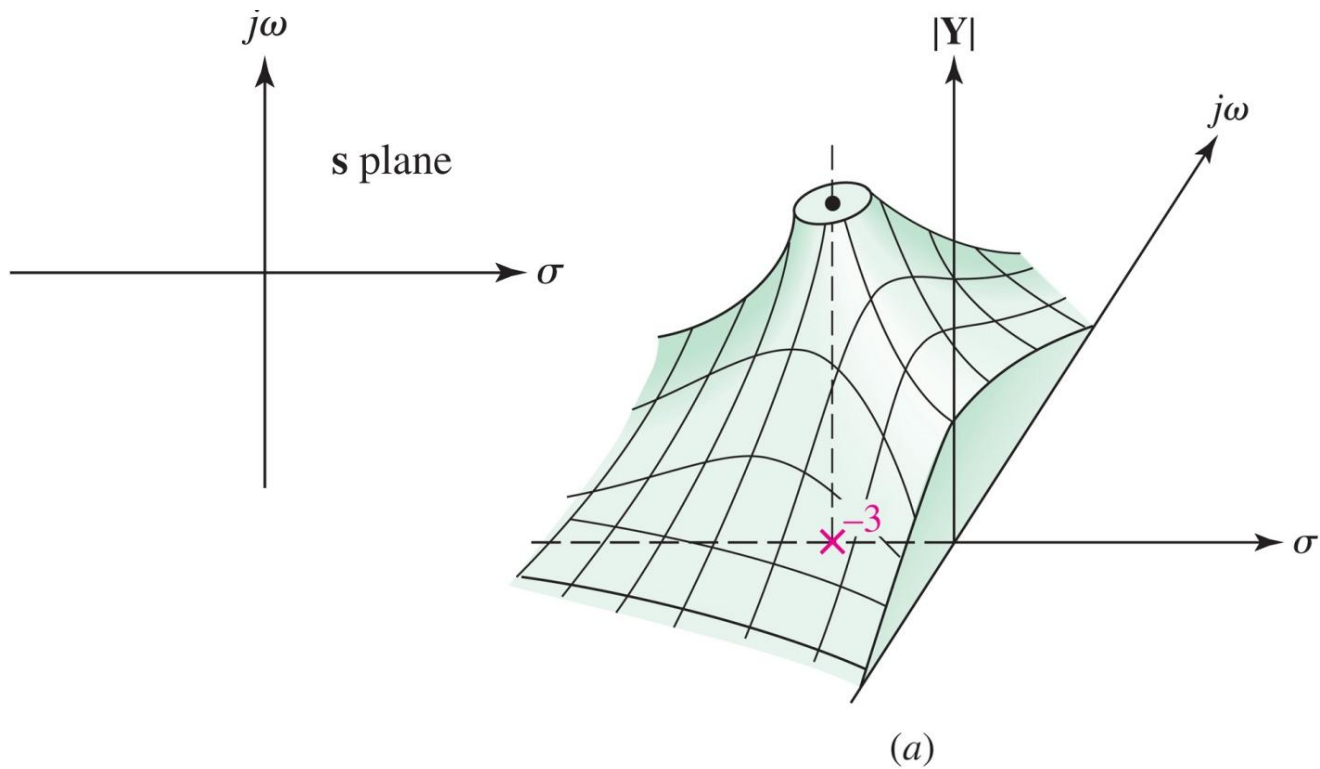
Zero of H(s)

$$s \rightarrow -1/(RC), H \rightarrow \infty$$

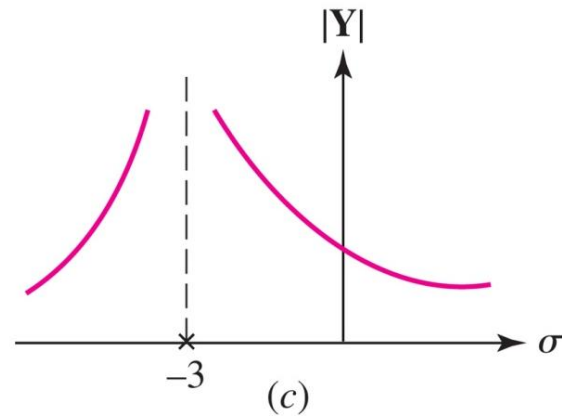
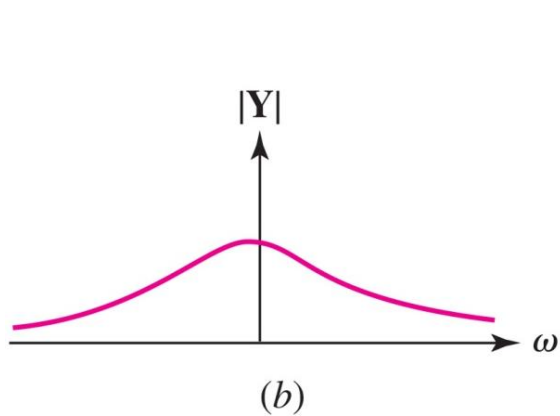
Pole of H(s)

These are **critical frequencies**

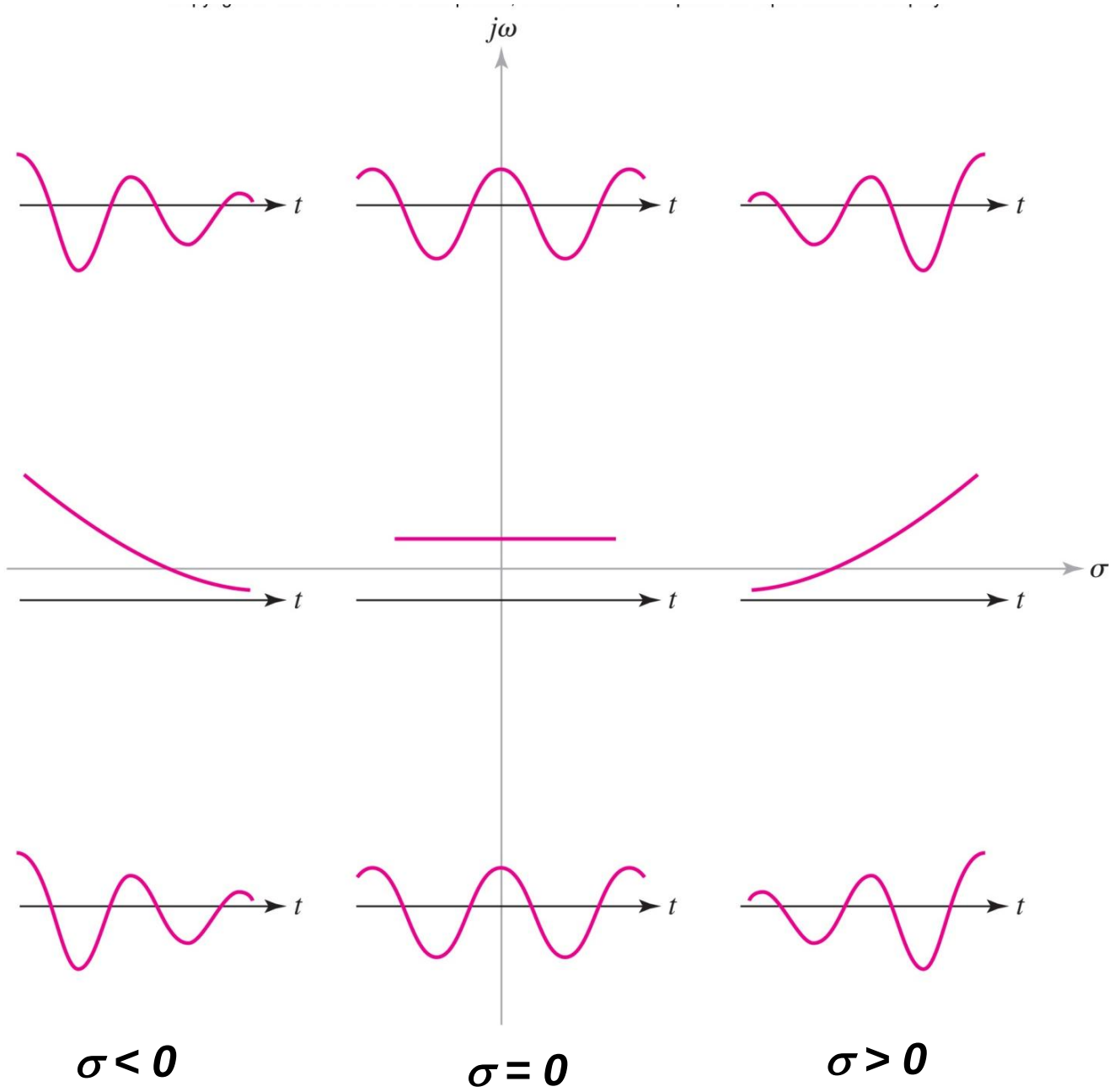
Complex-Frequency plane



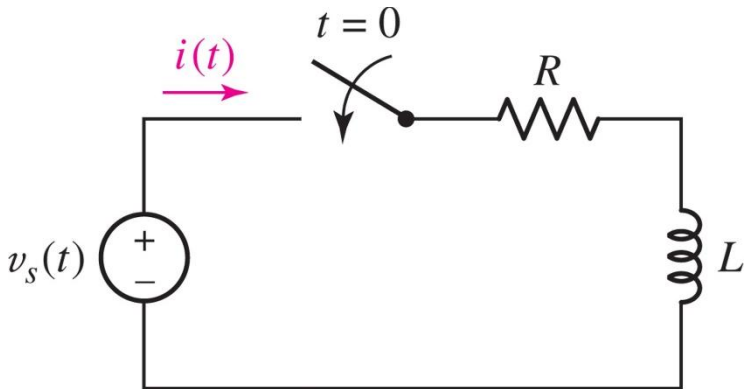
$$Y(s) = \frac{1}{s+3}$$



$$\omega = 0$$



Natural response and the s-plane



$$i(t) = i_n(t) + i_f(t)$$

Forced response:

$$I(s) = \frac{V_s}{R + sL} \Rightarrow i(t) = L^{-1}(I(s)) = i_n(t) + i_f(t)$$

Natural response:

$$H(s) = \frac{I}{V_s} = \frac{1}{L(s + R/L)} = Y(s)$$

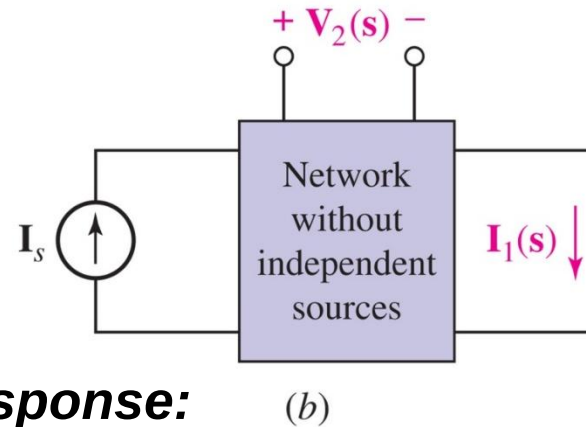
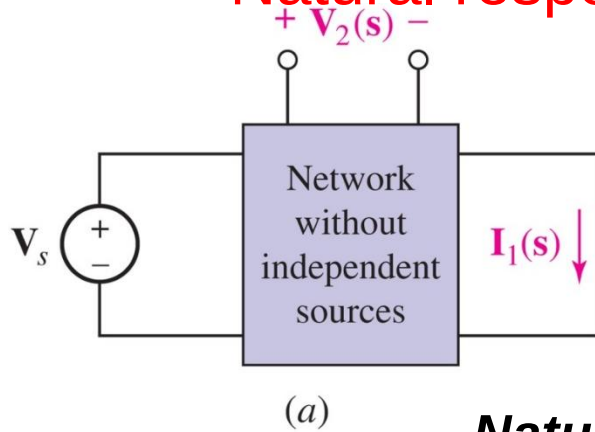
Pole is at $s = -R/L$

$$i_n(t) = Ae^{st} = Ae^{-\frac{R}{L}t}$$

A should be found from initial conditions

$$i(t) = Ae^{-\frac{R}{L}t} + i_f(t)$$

Natural response and the s-plane



Natural response:

For a):

$$H(s) = \frac{V_2(s)}{V_s} = \frac{(s - s_1)(s - s_3)\dots}{(s - s_2)(s - s_4)\dots}$$

Poles are at $s = s_2, s_4, \dots$

$$v_{2n}(t) = A_2 e^{s_2 t} + A_4 e^{s_4 t} + \dots \quad A_n \text{ should be found from initial conditions}$$

$$H(s) = \frac{I_1(s)}{V_s} = \frac{(s - s_1)(s - s_3)\dots}{(s - s_2)(s - s_4)\dots} \Rightarrow i_{1n}(t) = A_2 e^{s_2 t} + A_4 e^{s_4 t} + \dots$$

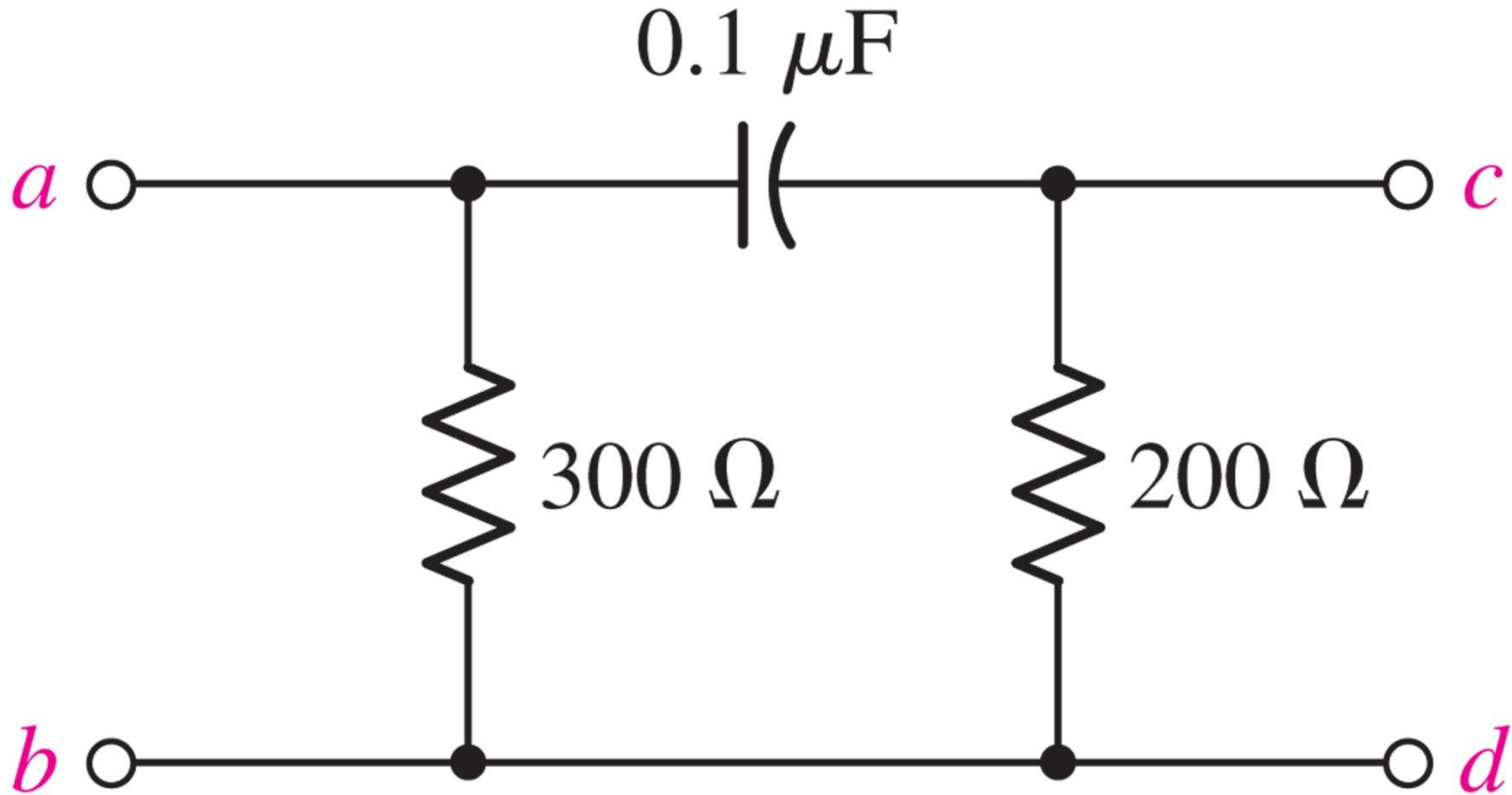
For b):

$$H(s) = \frac{I_1(s)}{I_s} = \frac{(s - s_1)(s - s_3)\dots}{(s - s_2)(s - s_4)\dots} \Rightarrow i_{1n}(t) = A_2 e^{s_2 t} + A_4 e^{s_4 t} + \dots$$

$$H(s) = \frac{V_2(s)}{I_s} = \frac{(s - s_1)(s - s_3)\dots}{(s - s_2)(s - s_4)\dots} \Rightarrow v_{2n}(t) = A_2 e^{s_2 t} + A_4 e^{s_4 t} + \dots$$

If a current source $i_1(t) = u(t)$ is present at a-b with the arrow entering a, find $H(s) = V_{cd} / I_1$, and specify the natural frequencies present in $v_{cd}(t)$.

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$$120s/(s+20000)\ \text{Ohm}; -20000\ \text{Hz}$$